Lecture 2
Basic Concepts and Model Evaluation

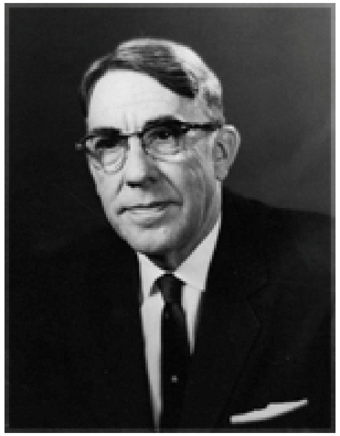
Lin ZHANG, PhD
School of Computer Science and Technology
Tongji University
Fall 2026

傍晚，小街路面上沁出微雨后的湿润，和煦的细风吹来，抬头看看天边的晚霞，嗯，明天又是一个好天气。走到水果摊旁，挑了个根蒂蜷缩、敲起来声音浊响的青绿西瓜，一边满心期待着皮薄肉厚瓤甜的爽落感，一边愉快地想着：这学期狠下了功夫，基础概念弄得清清楚楚，算法作业也是信手拈来，这门课成绩一定差不了！

摘自《机器学习》（周志华著，2016）

What is machine learning?

- Gives "computers the ability to learn without being explicitly programmed” (Arthur Samuel in 1959)



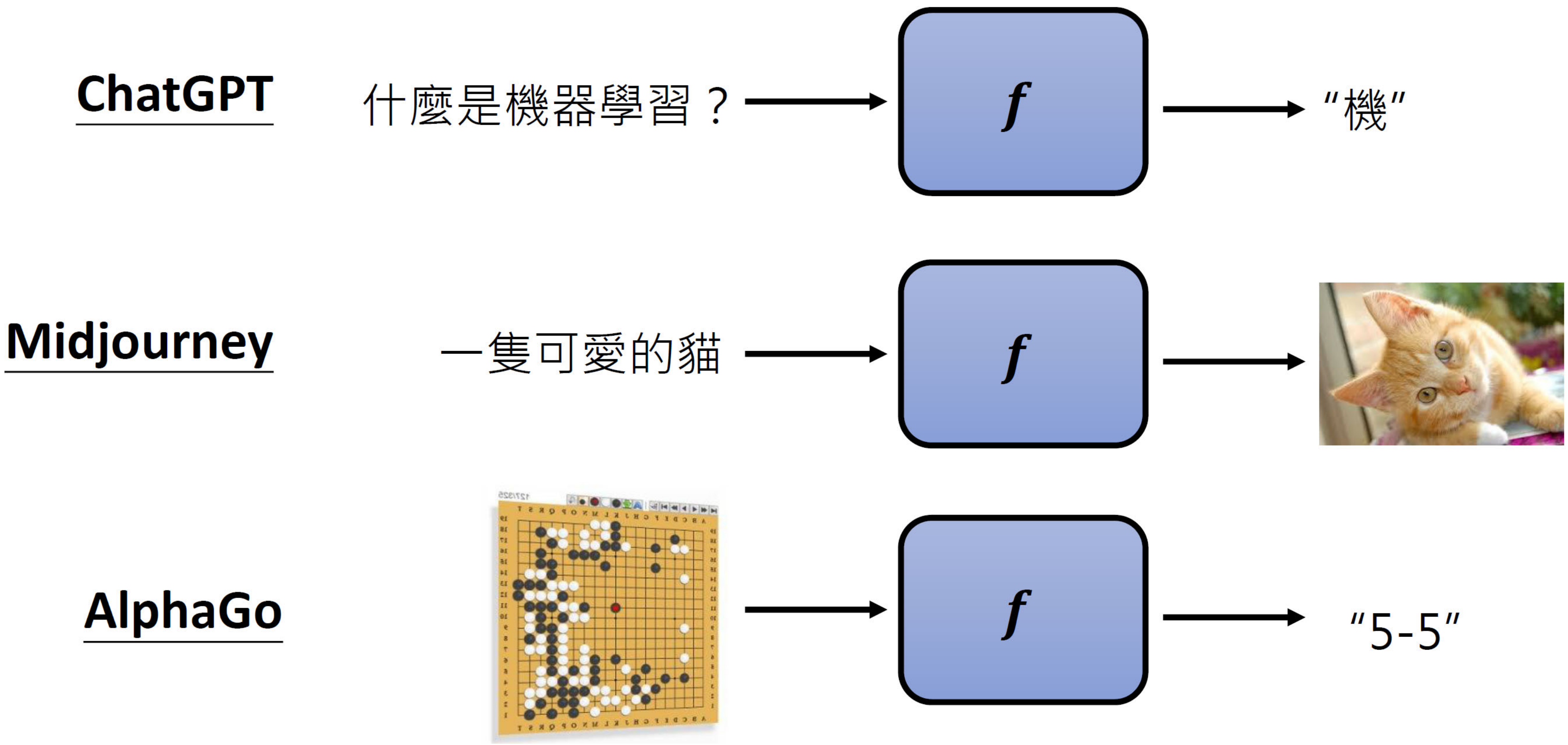
Arthur Lee Samuel
(December 5, 1901 – July 29, 1990)

- It explores the study and construction of algorithms that can learn from and make predictions on data
- It is employed in a range of computing tasks where designing and programming explicit algorithms with good performance is difficult or unfeasible

[1] Samuel, Arthur L., Some Studies in Machine Learning Using the Game of Checkers, IBM Journal of Research and Development, 1959

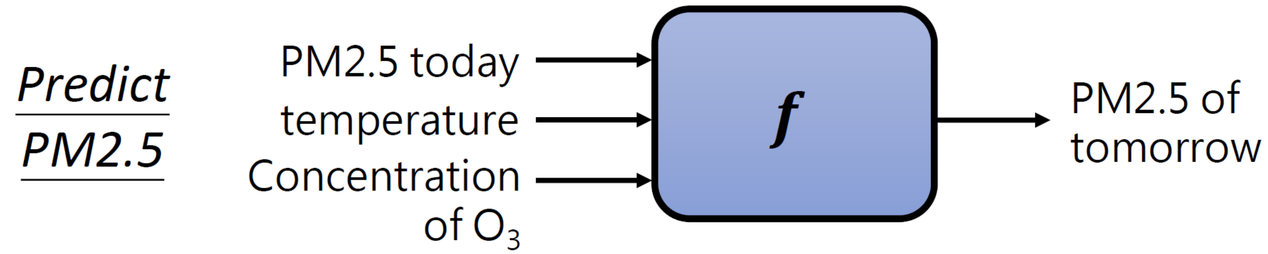
What is machine learning?

- The core goal of machine learning is to find a function f

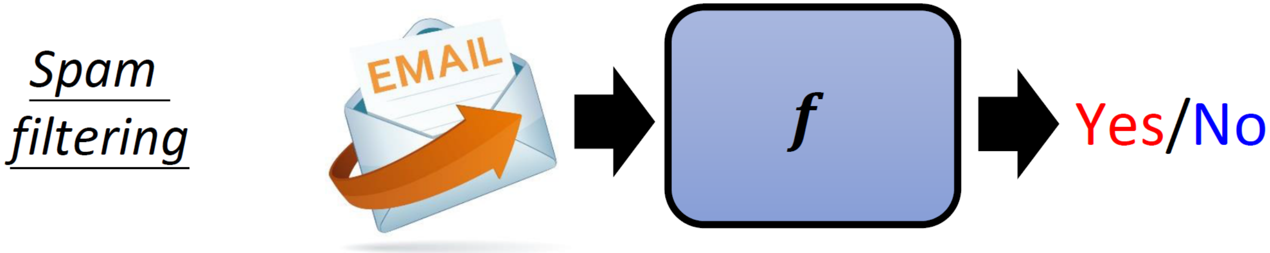


What is machine learning?

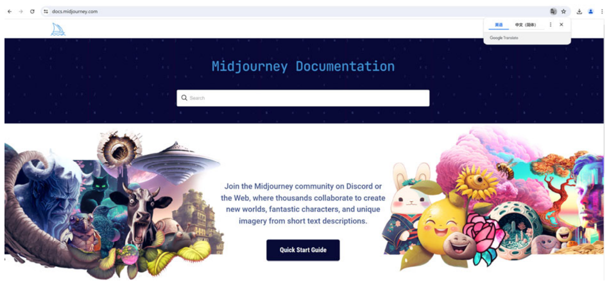
- According to the output, the machine learning problems can be categories as classification, regression and generative learning
  - Regression: the output of the function is a number



  - Classification: the output of the function is a choice from a fixed set



  - Generative learning: the output is a structured object (e.g., an image, a sentence)



What is machine learning?

- Three steps to identify f

Determine the function set	Specify the set of the functions Deep learning (CNN, RNN, Transformer), SVM (hyperplanes), Decision tree, etc.	Model
Determine the criteria	How to judge whether a function is “good” or “bad”? Supervised learning, Semi-supervised learning, Reinforcement learning, etc.	Loss
Figure out the “best” function	According to the criteria, how to identify the ‘best’ function from the set? Gradient descent, convex optimization, generic algorithm, etc.	optimization

About sample

- Attribute (feature), attribute value, label, and example

色泽，根蒂，敲声 → {好瓜，坏瓜}

features　　　　labels

{青绿，蜷缩，浊响：好瓜}

feature values　　label value

one example

Training, testing, and validation

- Training sample and training set

A training set comprising m training samples,

$$D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$$

where $\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{id}) \in \boldsymbol{\chi}$ is the feature vector of ith sample and $y_i \in \zeta$ is its label

By training, our aim is to find a mapping,

$$f : \boldsymbol{\chi} \mapsto \zeta$$

based on D

If ζ comprises discrete values, such a prediction task is called "**classification**"; if it comprises real numbers, such a prediction task is called "**regression**"

Training, testing, and validation

- Training sample and training set
- Test set
  - A test set is a set of data that is independent of the training data, but that follows the same probability distribution as the training data
  - Used only to assess the performance of a fully specified classifier

Training, testing, and validation

- Training sample and training set
- Test set
- Validation set
  - In order to avoid overfitting, when any classification parameter needs to be adjusted, it is necessary to have a validation set; it is used for model selection
  - The training set is used to train the candidate algorithms, while the validation set is used to compare their performances and decide which one to take

Overfitting, Generalization, and Capacity

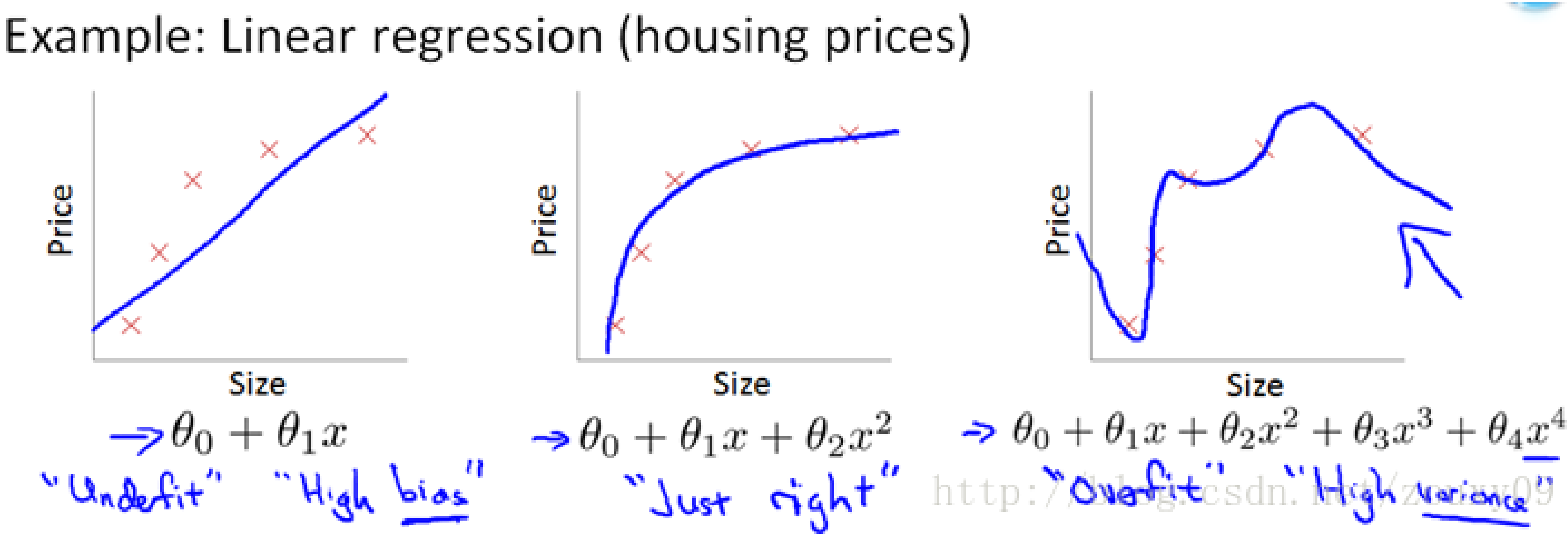
- Overfitting
  - It occurs when a statistical model describes random error or noise instead of the underlying relationship
  - It generally occurs when a model is excessively complex, such as having too many parameters relative to the number of observations
  - A model that has been overfit will generally have poor predictive performance, as it can exaggerate minor fluctuations in the data

Overfitting, Generalization, and Capacity

- Overfitting
- Generalization
  - Refers to the performance of the learned model on new, previously unseen examples, such as the test set

Overfitting, Generalization, and Capacity

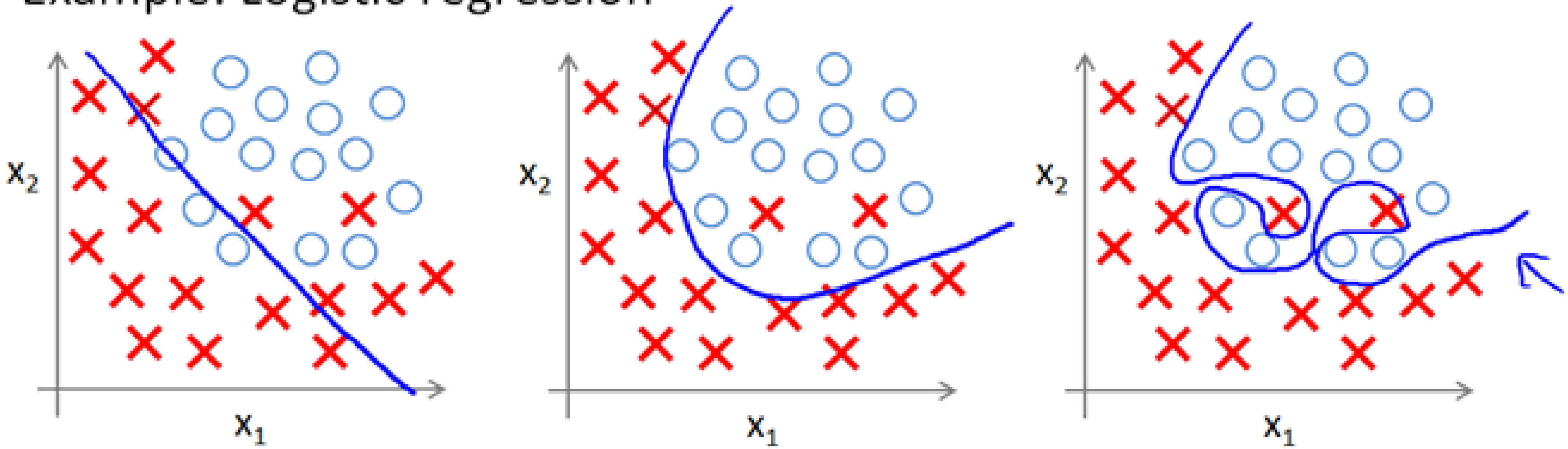
- Overfitting
- Generalization



Overfitting, Generalization, and Capacity

- Overfitting
- Generalization

Example: Logistic regression



$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

(g = sigmoid function)

"Underfit"

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2)$$

$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \theta_6 x_1^3 x_2 + \dots)$$

"Overfit"

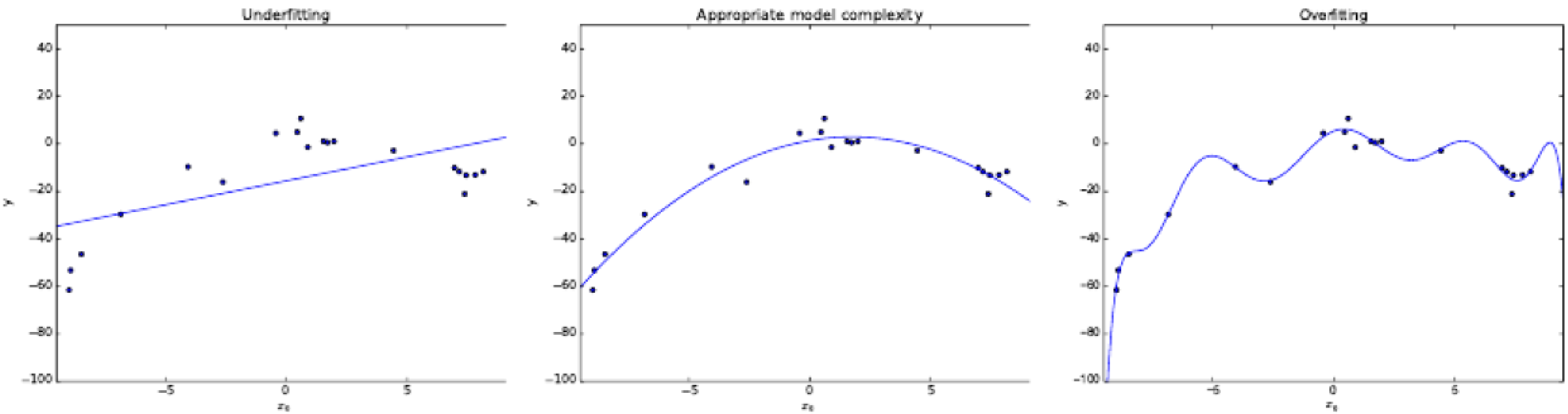
Overfitting, Generalization, and Capacity

- Overfitting
- Generalization
- Capacity
  - Measures the complexity, expressive power, richness, or flexibility of a classification algorithm
  - Ex, DCNN (deep convolutional neural networks) is powerful since its capacity is very large

$$y^* = b + \omega x, \quad y^* = b + \omega_1 x_1 + \omega_2 x_2, \quad y^* = b + \sum_{i=1}^{10} \omega_i x_i$$

$\longrightarrow$ higher capacity

Overfitting, Generalization, and Capacity



higher capacity

Performance Evaluation

Given a sample set (training, validation, or test)

$$D = \{(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_m, y_m)\}$$

To assess the performance of the learner f, we need to compare the prediction $f(\boldsymbol{x})$ and its ground-truth label y

For regression task, the most common performance measure is MSE (mean squared error),

$$E(f; D) = \frac{1}{m} \sum_{i=1}^{m} \left(f(\boldsymbol{x}_i) - y_i \right)^2$$

Performance Evaluation (for classification)

- Error rate
  - The ratio of the number of misclassified samples to the total number of samples

$$E(f;D)=\frac{1}{m}\sum_{i=1}^{m}\mathbf{1}\left(f(\boldsymbol{x}_i)\neq y_i\right)$$

- Accuracy
  - It is derived from the error rate

$$acc(f;D)=\frac{1}{m}\sum_{i=1}^{m}\mathbf{1}\left(f(\boldsymbol{x}_i)=y_i\right)=1-E(f;D)$$

Performance Evaluation (for classification)

- Precision and Recall

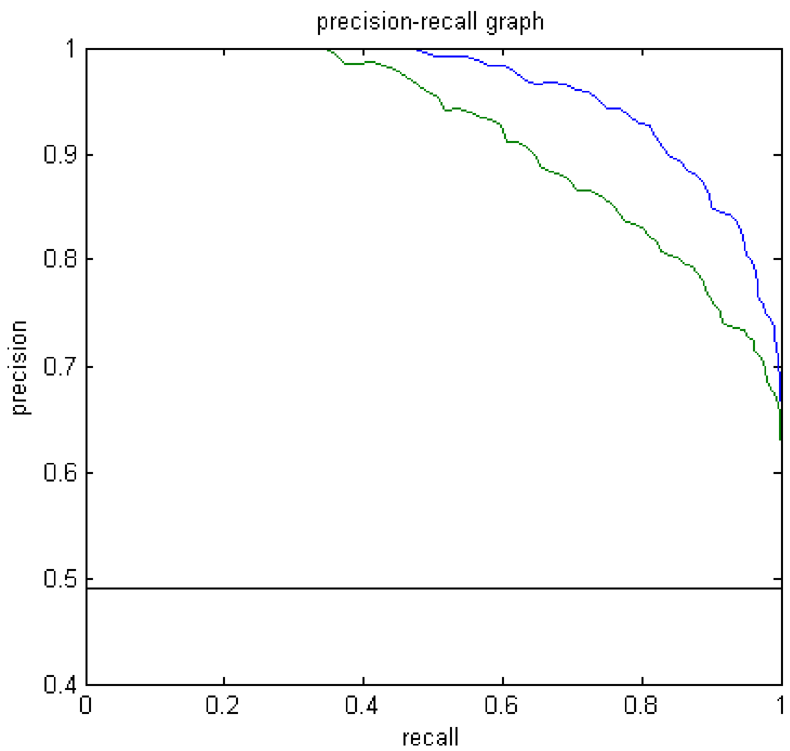
Ground truth	Prediction	
	positive	negative
positive	True Positive (TP)	False Negative (FN)
negative	False Positive (FP)	True Negative (TN)

$$precision = \frac{TP}{TP + FP}$$

$$recall = \frac{TP}{TP + FN}$$

Performance Evaluation (for classification)

- Precision and Recall
  - Often, there is an inverse relationship between precision and recall, where it is possible to increase one at the cost of reducing the other
  - Usually, PR-curve is not monotonic



Performance Evaluation (for classification)

- Precision-recall should be used together; it is meaningless to use only one of them
- However, in many cases, people want to know explicitly which algorithm is better; we can use F-measure

$$F_\beta = \frac{(1+\beta^2)\times P\times R}{(\beta^2\times P)+R}$$

Performance Evaluation (for classification)

- To derive a single performance measure

Varying threshold, we can have a series of (P, R) pairs,

$$(P_1, R_1), (P_2, R_2), \ldots, (P_n, R_n)$$

Then,

$$P_{macro} = \frac{1}{n}\sum_{i=1}^{n} P_i \qquad R_{macro} = \frac{1}{n}\sum_{i=1}^{n} R_i$$

$$F_{\beta-macro} = \frac{\left(1+\beta^2\right)\times P_{macro}\times R_{macro}}{\left(\beta^2\times P_{macro}\right)+R_{macro}}$$

Model selection—Cross validation

- Simple cross validation
  - Split the dataset at hand into a training set and a validation set
  - Training the models on the training set, and selecting the best model based on the evaluation on the validation set
- S-fold cross validation
  - Randomly split the dataset at hand into S equal-sized subsets; any two subsets do not overlap with each other
  - For one learning model, train it on S-1 subsets and evaluate its performance on the remaining one subset; repeat such a training-evaluating procedure S times, each time using a different subset for evaluation; averaging the obtained S evaluation errors as the performance of this learning model
- Leave-one-out cross validation
  - It can be regarded as a special case of the S-fold cross validation strategy, i.e., S=m, m is the number of training samples

Class-imbalance Issue

- Problem definition
  - It is the problem in machine learning where the total number of a class of data is far less than the total number of another class of data
  - This problem is extremely common in practice
- Why is it a problem?
  - Most machine learning algorithms work best when the number of instances of each classes are roughly equal
  - When the number of instances of one class far exceeds the other, problems arise

Class-imbalance Issue

- How to deal with this issue?
  - Modify the cost function
  - Under-sampling, throwing out samples from majority classes
  - Oversampling, creating new virtual samples for minority classes
    - » Just duplicating the minority classes could lead the classifier to overfitting to a few examples
    - » Instead, use some algorithm for oversampling, such as SMOTE (synthetic minority over-sampling techniqe)[1]

[1] N.V. Chawla *et al.*, SMOTE: Synthetic Minority Over-sampling Technique, J. Artificial Intelligence Research 16: 321-357, 2002

Class-imbalance Issue

- Minority oversampling by SMOTE[2]

Add new minority class instances by:

- For each minority class instance c
  - neighbours = Get KNN(5)
  - n = Random pick one from neighbours
  - Create a new minority class r instance using c's feature vector and the feature vector's difference of n and c multiplied by a random number
    - » i.e. r.feats = c.feats + (n.feats - c.feats) * rand(0,1)

[2] N.V. Chawla *et al.*, SMOTE: Synthetic Minority Over-sampling Technique, J. Artificial Intelligence Research 16: 321-357, 2002


Q+A